

Molecular diversity and tools for deciphering the methanogen community structure and diversity in freshwater sediments

Prem Prashant Chaudhary · Lenka Brablcová ·
Iva Buriánková · Martin Rulík

Received: 17 June 2013 / Revised: 5 July 2013 / Accepted: 5 July 2013
© Springer-Verlag Berlin Heidelberg 2013

Abstract Methanogenic archaeal communities existing in freshwater sediments are responsible for approximately 50 % of the total global emission of methane. This process contributes significantly to global warming and, hence, necessitates interventional control measures to limit its emission. Unfortunately, the diversity and functional interactions of methanogenic populations occurring in these habitats are yet to be fully characterized. Considering several disadvantages of conventional culture-based methodologies, in recent years, impetus is given to molecular biology approaches to determine the community structure of freshwater sedimentary methanogenic archaea. 16S rRNA and methyl coenzyme M reductase (*mcrA*) gene-based cloning techniques are the first choice for this purpose. In addition, electrophoresis-based (denaturing gradient gel electrophoresis, temperature gradient gel electrophoresis, and terminal restriction fragment length polymorphism) and quantitative real-time polymerase chain reaction techniques have also found extensive applications. These techniques are highly sensitive, rapid, and reliable as compared to traditional culture-dependent approaches. Molecular diversity studies revealed the dominance of the orders *Methanomicrobiales* and *Methanosarcinales* of methanogens in freshwater sediments. The present review discusses in detail the status of the diversity of methanogens and the molecular approaches applied in this area of research.

Keywords Methanogens · Diversity · Sediments · Microbial diversity · Ribosomal RNA and *mcrA*

Introduction

Microbial communities inhabiting anoxic environments like freshwater sediments, marine sediments, and shallow regions of the sea are considered to be very important in the cycling of carbon and sulfur (Nedwell 1984). Methanogenic archaea are considered to play an important role in the mineralization of organic carbon to methane (CH₄) (Capone and Kiene 1988; DeLong 1992). Methane is an important greenhouse gas present at a concentration of 1.8 ppmv (approximately) and contributes about 15 % in the anthropogenic greenhouse effect (Forster et al. 2007). Methane concentration in the atmosphere has been dramatically increased due to industrial revolution and urbanization from 0.7 to 1.8 ppmv from 1999 to 2005 (Forster et al. 2007). This unexpected increase in the concentration of methane has been published by Rigby et al. (2008), suggesting that both sources and sinks of atmospheric methane is dynamic and evolving but is not very well understood. Freshwater sediments like wetlands, rice paddy fields, and lakes play a very important role in this annual atmospheric flux and contribute about 40–50 % of the annual methane emission (Cicerone and Oremland 1988; Conrad 2009; Rulík et al. 2013). Methanogenesis is the final step in the degradation of organic matter in anoxic environments, and hence, methane is introduced into the environment. So, it is worthwhile to study the diversity and ecology of methanogens in these freshwater sediments in order to understand methane emission from these sources which further leads to global warming (Ehhalt and Prather 2001). Information about the distribution and diversity of methanogens present inside freshwater sediments is scanty. Traditional approaches used to count and identify methanogens from these sediments by direct culturing and microscopy are very time-consuming and require a high degree of accuracy for correct identification.

It has been observed that, normally, there are large amounts of differences in the diversity studies carried out

P. P. Chaudhary (✉) · L. Brablcová · I. Buriánková · M. Rulík
Laboratory of Aquatic Microbial Ecology, Department of Ecology
and Environmental Sciences, Faculty of Science, Palacký
University, Olomouc, Czech Republic
e-mail: luckypgi@gmail.com

using culture-dependent and culture-independent techniques and the latter gave us an idea about the existence of microbes which requires stringent environmental conditions to grow. These factors along with their high sensitivity, competence, and amenability make these culture-independent techniques the first choice of the investigators to decipher the diversity of microbes in the natural environment (Nocker et al. 2007). However, culture-based methods are very important in the context of obtaining the pure isolates of the novel microbes and the long-term storage of the reference strains for future usage. This review describes the current status of methanogen diversity present in freshwater sediments and also tries to give an outlook of the molecular approaches used by the investigators to study the diversity of methanogens from these sources by now. To our knowledge, this is the first review to deal with the molecular techniques used to investigate the diversity of methanogens in sediments as most of the reviews are published in the area of ruminant sciences and with focus on only one or two techniques (Sirohi et al. 2012; Kumar et al. 2011). The first report of the use of molecular techniques in microbial ecology was published in 1990. In this study, researchers directly amplified 16S rRNA gene from Sargasso Sea bacterioplankton and then sequenced the amplified product (Giovannoni et al. 1990). Molecular methods since then have been widely used and applied to study the microbial communities in different habitats including methanogen communities in freshwater sediments (Watanabe et al. 2004; Ganzert et al. 2007; Orphan et al. 2008; Cetecioglu et al. 2009; Jiang et al. 2011; Lazar et al. 2012).

Methanogens

Methanogens are obligate anaerobes belonging to the phylum Euryarchaeota and comes under the domain archaea (Woese et al. 1990; Zhou et al. 2011). Methanogens are broadly classified into four groups according to the substrate used: (1) genera using H_2/CO_2 and formate, e.g., *Methanobacterium*, *Methanobrevibacter*, and *Methanogenium*; (2) genera using acetate, e.g., *Methanosaeta*; (3) genera using acetate, H_2/CO_2 , and methyl compounds, e.g., *Methanosarcina*; and (4) genera using methyl compounds such as methanol, e.g., *Methanolobus* and *Methanococcus* (Garcia 1990). Methanogens as a unique characteristic feature have coenzyme F420 which will further act as cofactor for enzymes involved in methane production like formate dehydrogenase and hydrogenase (Sharma et al. 2011). Also, in addition, the methyl coenzyme reductase enzyme complex is thought to be inimitable in methanogens, making this a marker gene for their detection. The *mcrA* form of this enzyme complex is thought to be present in all methanogen types, while the presence of the *mcr-II* form has only been restricted to the members of the orders *Methanobacteriales* and

Methanococcales (Chaudhary et al. 2011). This gene is thought to catalyze the reduction of the methyl group bound to coenzyme M, with the associated release of methane. This gene, along with the 16S rRNA gene, has been widely used by researchers to decipher the diversity of methanogens from rumen and freshwater sediments and many other extreme environments (Chaudhary 2009; Chaudhary and Sirohi 2009; Chaudhary et al. 2011, 2012; Ganzert et al. 2007; Cahyani et al. 2004; Zeleke et al. 2013; Rastogi et al. 2011; Sirohi et al. 2013).

Role of methanogens in global warming

Methane is a potent greenhouse gas present in the earth's atmosphere and has about 25 % more global warming potential than CO_2 . Therefore, methanogenesis in ruminants and decay of organic matter deep inside the earth are supposed to be the major contributors in the problem of global warming, although deep inside the earth we cannot consider methane as a direct contributor as it uses atmospheric carbon dioxide when it is created and works on organic matter. But at the end, it converts CO_2 into CH_4 which is a much more potent greenhouse gas (Young 1992). The rate of methanogenesis in sediments is controlled by the relative availability of substrates, mainly acetate, H_2 , and CO_2 . Methanogens occur through both ways in sediments, i.e., aerobically as well as anaerobically, and are controlled by different groups of methanogenic archaea. Deep inside the sediments, methane is produced anaerobically by methanogens using acetate, H_2 , and CO_2 as their substrate. However, the methane produced by this process is further oxidized by aerobic methane-oxidizing bacteria, i.e., methanotrophs. Thus, the overall release of methane in the environment depends upon the availability of these methanotrophs in the sediments as well (Conrad 2009). In spite of this barrier which inhibits the direct emission of methane into the environment, it has been estimated that methane emission from sediments contributes about 40–50 % of the total methane emission in the environment.

Diversity and distribution of methanogens in sediments

The diversity and distribution of methane have been found to vary according to the depth, type of substrate available, and also temperature variations. Earl et al. (2005) studied two sediment samples from Priest Pot, a small (1 ha), shallow hypereutrophic lake in the English Lake District, at different time intervals from October 1999 until October 2000 and showed that methanogen diversity remains same in both sediment samples obtained at depths of 0, 1, 2, 5, 10, and 14 cm, but they found that temperature variation has a

profound effect on the diversity of methanogens. Diversity seems to be less from winter through spring and summer but increased in the season of autumn. Nolla-Ardèvol et al. (2012) showed that the orders *Methanobacteriales* and *Methanomicrobiales* of methanogens are dominant in Asian soda lakes. Putkinen et al. (2009), in their study on samples from peatland, showed that peatland depths have a clear-cut effect on community structure. Samples isolated from the deepest depth vary remarkably, as one sample showed the presence of several methanogenic groups while the other have a high number of RC-II archaea-like clones which are thought to be methanogens based on their phylogenetic position between the *Methanosarcinales* and *Methanomicrobiales* groups (Grosskopf et al. 1998). These differences in community structure on the basis of site and depth might be due to the differences in hydrological conditions of the sites under study. In another study carried out in the Marmara Sea, sediment samples from seven different depths have been analyzed and all these seven points were polluted with domesticated and industrial discharges. Results showed that the diverse microbial communities are equally distributed at all the depths but the dominant order of methanogens depend significantly on the depth and S, Mn, and Fe content of the sediments (Cetecioglu et al. 2009). In freshwater sediments, as the level of salinity is low and sulfur content is less, so most of the time, methanogens of the order *Methanosarcinales* come out as an important methanogen group which utilizes acetate, hydrogen/carbon dioxide, or methyl compounds (e.g., methanol, methyl amines) as their substrate for the production of methane (Earl et al. 2005). *Methanobacterium* species were most abundant in sediments, although *Methanosarcina*, *Methanococcus*, and *Methanospirillum* species were observed in enrichment cultures. The most dominant methanogen populations had simple nutritional requirements, and the optimum temperature for their growth varies from 4 to 45 °C (Zeikus and Winfrey 1976).

Another study carried out on sediment samples isolated from Guerrero Negro, Baja California Sur at different depths revealed some interesting facts. It shows that, at the surface layer, *Methanlobus*-related methanogens were dominant. When they were studied at a depth of 100–155 mm, a shift in the dominant methanogen genera from *Methanlobus* to *Methanococcoides* occurs. This shift in the dominant methanogen phylotypes with an increase in depth might be because members of *Methanococcoides* represent an environment of relatively lower nutrients and carbon but greater stability as compared to the active surface layers of the mat (Orphan et al. 2008). Diversity analysis of methanogens have also been carried out on sediment samples (from six different depths) isolated from intertidal mudflat in the Yangtze River Estuary (China), and scientists observed that *Methanomicrobiales* and *Methanosarcinales* are the

dominant methanogen orders throughout the sediment layers, but a variation in their counts has been observed with an increase in depth. The dominance of the hydrogenotrophic methanogen *Methanomicrobiales* slightly increased with depth, while *Methanosarcinales* showed a reverse pattern (Zelege et al. 2013). Purdy et al. (2002) have also studied the differences in the diversity of methanogenic communities at the brackish and marine ends of a UK estuary. It has been found that the *Methanosarcina* and *Methanocorpusculum* genera were dominant, and more precisely, most of the clones resemble the acetate-utilizing *Methanosaeta concilii* which is omnipresent in freshwater sediments, while the phylotypes related to *Methanoculleus* and *Methanococcoides* were dominant in marine sediments. In another study carried out on sediment samples from Lonar Lake, Pune, India, Surakasi et al. (2007) have found that the *Methanoculleus* and *Methanocalculus* groups of methanogens are vastly found in these sediment samples, after the uncultured group of archaea, which is the dominant one. This diversity of methanogens is far more different from the methanogenic diversity found in other soda lakes and much more similar to those found in palm oil, oil reservoirs, paddy fields, and waste treatment digesters. Yoshioka et al. (2010) studied the diversity of methanogens in sediment samples isolated from different depths inside and outside the area of gas hydrate distribution on the Cascadia Margin. They found that the *Methanosarcinales* group of methanogens is dominant in near-surface sediments, while the *Methanobacteriales* group is the most abundant above the gas hydrate zone. In one of the studies carried out in The Netherlands, Lazar and his coworkers studied sediment samples (different depths) from the eastern Mediterranean Sea. They observed that methanogenic diversity increases with an increase in depth (up to 100–120 cm) but again decreases when the depth increases from this range as they found only one methanogenic band during denaturing gradient gel electrophoresis (DGGE) analysis in the case of sediments sample collected from the deepest core (600–650 cm). In this study, *Methanosarcinales*-affiliated sequences were detected using molecular approaches and enrichments of methanogens showed the presence of *Methanococcoides* in shallow sediment layers (Lazar et al. 2012). All these studies suggest that methanogen diversity in sediments differ to a great extent and mostly dependent upon the availability of substrates inside the sediments for the growth of any particular type of methanogenic strain. A shift in the temperature range also shows a profound effect on the diversity of methanogens in the sediments. A detailed description of the methanogen species or orders identified, clone library size or operational taxonomic units (OTUs), and molecular techniques employed is given in Table 1. It has been concluded from the studies that *Methanomicrobiales* is the dominant order of methanogens and constitutes about 43 % of the total methanogens found in sediments, while *Methanosarcinales* is the second most

Table 1 Molecular techniques for determining the population density and diversity of methanogens in sediments

Source	Target gene(s)	Molecular technique(s) used	Number of phylotypes/OTUs	Geographical region	Dominant species/ genera/order identified	References
Sediment samples from coastal salt marsh	16S rRNA (rDNA)	PCR, cloning, and sequencing	30	UK	<i>Methanococcoides</i> and <i>Methanolobus</i>	Munson et al. (1997)
Rice roots and anoxic bulk soil of flooded rice microcosms	SSU rRNA	PCR, cloning, and sequencing	32	Germany	<i>Methanosarcina</i> spp./ <i>Methanobacterium</i> spp.	Grosskopf et al. (1998)
Anoxic sediments from Rotsee	16S rRNA	PCR, cloning, sequencing, and FISH	17	Switzerland	<i>Methanosarcina</i> spp.	Zepp Falz et al. (1999)
Anoxic paddy soil	SSU rRNA	T-RFLP	60	Germany	<i>Methanobacteriaceae</i> / <i>Methanobacterium</i> spp.	Weber et al. (2001)
Sediments from brackish and marine ends of a UK estuary	16S rRNA (rDNA)	PCR, cloning, and sequencing	12	UK	<i>Methanosarcina</i> spp.	Purdy et al. (2002)
Nodal roots of rice plants in flooded soil pots	16S rRNA	PCR-DGGE and sequencing	42	Japan	<i>Methanomicrobiales</i>	Ikenaga et al. (2004)
Paddy field soil	16S rRNA	PCR-DGGE and sequencing	41	Japan	<i>Methanomicrobiales</i>	Watanabe et al. (2004)
Rice straw from rice field	16S rRNA	PCR-DGGE, cloning, and sequencing	12	Japan	<i>Methanomicrobiales</i>	Cahyani et al. (2004)
Sediment samples from Bor	16S rRNA	PCR, cloning, RFLP, and sequencing	25	Thailand	na	Kanokratana et al. (2004)
Khlung hot spring	16S rRNA	PCR, cloning, TTGE, and sequencing	na	UK	<i>Methanosarcina</i> spp. and <i>Methanomicrobiales</i>	Earl et al. (2005)
Sediments samples from small eutrophic lake, Priest Pot in the English Lake District	<i>mcrA</i> and 16S rRNA	PCR, cloning, and sequencing	13 (<i>mcrA</i>), 22 (16S rRNA)	USA	<i>Methanomicrobiales</i>	Dhillon et al. (2005)
Hydrothermal sediments of Guaymas Basin	16S rRNA	PCR-DGGE, cloning, and sequencing	54	Japan	<i>Methanomicrobiales</i> and <i>Methanosarcinales</i>	Wani et al. (2006)
Paddy field soil under double cropping	16S rRNA	PCR, cloning, RFLP, and sequencing	13	India	na	Karr et al. (2006)
Upper sediments from Lonar soda lake	16S rRNA	PPCR, DGGE, and sequencing	na	Antarctic	<i>Methanosarcinales</i> and <i>Methanoculleus</i>	
Sediment samples from Lake Fryxell	16S rRNA	PCR, cloning, sequencing, and RFLP	15	India	<i>Methanoculleus</i> spp. and <i>Methanocaldococcus</i> spp.	Ganzert et al. (2007)
Sediments of Lonar Lake	16S rRNA	PCR, cloning, sequencing, and T-RFLP	na	USA	<i>Methanogenicum</i>	Goffredi et al. (2008)
Permafrost soils	16S rRNA	PCR, cloning, and sequencing	na	USA	<i>Methanosarcinales</i>	Orphan et al. (2008)
Sediment samples from a whale fall in Monterey Canyon, California	16S rRNA and <i>mcrA</i>	PCR, cloning, and sequencing	na	Turkey	Uncultured archaeon	Cetecioglu et al. (2009)
Sediment samples from Guerrero Negro, Baja California Sur, Mexico	16S rRNA	PCR, DGGE, and sequencing	18	Finland	Rice cluster II (RC-II) and <i>Methanosarcinaceae</i>	Putkinen et al. (2009)
Sediment samples from Marmara Sea	16S rRNA	Reverse transcriptase PCR, T-RFLP, cloning, RFLP, and sequencing	na			
Sediment samples from two Finnish fen-type peatlands	16S rRNA					

Table 1 (continued)

Source	Target gene(s)	Molecular technique(s) used	Number of phylotypes/OTUs	Geographical region	Dominant species/ genera/order identified	References
Metal-contaminated Coeur d'Alene River sediments	<i>mcrA</i>	PCR, cloning, and sequencing	3	USA	<i>Methanosarcinales</i>	Rastogi et al. (2011)
Water column and sediment samples from lake Pavin	<i>mcrA</i>	PCR cloning and sequencing	20	France	<i>Methanomicrobiales</i>	Bidierre-Petit et al. (2011)
Sediment samples from Nyegga region of Norwegian Sea	16S rRNA, <i>mcrA</i> , and <i>dsrB</i>	PCR, DGGE, fingerprinting analysis, band excision, and sequencing	na	Norway	<i>Methanosarcinales</i> and <i>ANME-2</i>	Lazar et al. (2011)
Shallow sediments of the Pearl River Estuary	<i>mcrA</i>	PCR, cloning, and sequencing	na	China	<i>Methanomicrobiales</i>	Jiang et al. (2011)
Sediment samples from East Mediterranean Sea	16S rRNA and <i>mcrA</i>	PCR, DGGE fingerprinting analysis, band excision, and sequencing	16	Mediterranean Sea	<i>Methanosarcinales</i> and <i>ANME-2</i>	Lazar et al. (2012)
Sediments of Lonar Lake	16S rRNA and <i>mcrA</i>	PCR, cloning, and sequencing	4 (16S rRNA) and 8 (<i>mcrA</i>)	India	<i>Methanocaldococcus</i> sp. (in case of 16S rRNA), <i>Methanolinea</i> sp. (in case of <i>mcrA</i>)	Antony et al. (2012)
Sediments from the Central Asian hypersaline soda lakes	<i>mcrA</i>	PCR, cloning, and sequencing	7	Russia	<i>Methanobacterium</i> and <i>Methanothermobacter</i>	Nolla-Ardévol et al. (2012)
Anoxic sediments of meromictic Lake Kivu	16S rRNA	PCR, cloning, sequencing, and T-RFLP	50	Switzerland	Uncultured archaeon	Bhattarai et al. (2012)
Mudflat sediments of Yangtze River Estuary	<i>mcrA</i>	454-pyrosequencing and quantitative real-time PCR	na	China	<i>Methanosarcinales</i> and <i>Methanomicrobiales</i>	Zelege et al. (2013)

This table also includes the studies from marine sediment samples and is, therefore, not strictly limited to freshwater sediments only
na not available

dominant order of methanogens and comprises about 39 % of the total methanogens population and *Methanobacteriales*, uncultured archaeon, and *Methanococcales* contribute about 9, 6, and 3 %, respectively, of the methanogen population in sediments (Fig. 1).

Molecular techniques used to decipher the diversity of methanogens in sediments

16S or *mcrA* cloning and sequencing

Most of the diversity studies in sediments have been done using the most common molecular approach based on the polymerase chain reaction (PCR) amplification of 16S rRNA from community DNA, followed by cloning and sequencing. The sequences obtained undergo some bioinformatic analyses and result in the preparation of a phylogenetic tree which gives an idea about the diversity of microbes in environmental samples. The number of phylotypes or OTUs also suggests the abundance of any particular type of microbe in an environmental sample (Table 1). 16S or *mcrA* gene-based approaches are the most widely used approaches to study the diversity in sediments (Munson et al. 1997; Zepp Falz et al. 1999; Purdy et al. 2002; Dhillon et al. 2005; Antony et al. 2012). *mcrA* gene-based analysis has also been performed to study the methanogenic diversity in sediments (Biderre-Petit et al. 2011; Jiang et al. 2011; Rastogi et al. 2011; Nolla-Ardèvol et al. 2012a, b), although this method of diversity analysis has its own limitations as well. The first and most important limitation is the method adopted for community DNA isolation as it should be the one which will isolate the DNA of most of the microbial communities present in the target sample or we can choose the method by taking an account of the cell wall composition of the microbes under study. Another limitation of this technique is that, in order to cover the maximum diversity, one should study a large number of clones. This process is much more time-consuming and

expensive, too (Zoetendal et al. 2004). Nowadays, the fingerprinting technique has also been coupled with this method to reduce the cost of sequencing as it will help in the screening of clones with similar sequences and, hence, only a very limited number of clones need to be sequenced (Kanokratana et al. 2004; Wani et al. 2006).

Denaturing gradient gel electrophoresis

The first group to determine the role of DGGE in studying the soil microbial ecosystem was that of Muyzer et al. (1993). Since then, a number of scientists have adopted this method to study microbial diversity in different environmental samples, including sediments (Keyser et al. 2006). DGGE also involves the use of PCR-amplified 16S rRNA genes. In the DGGE process, amplified fragments have been separated from each other based on the slow movement of the partially melted dsDNA band in the gel which was formed with a linear gradient of urea and formamide (Muyzer et al. 1993; Muyzer and Smalla 1998). It has been estimated that this technique is useful for detecting the microbial communities which constitute only about 1 % of the total microbial population. DGGE has also been widely used for detecting methanogenic communities and other microbial communities in sediments (Watanabe et al. 2004; Ikenaga et al. 2004; Keyser et al. 2006; Ganzert et al. 2007). This way of elucidating diversity has several advantages over the PCR cloning technique as it will give an immediate picture of the number of microbial communities inhabiting in a given sample and in both the quantitative and semiquantitative ways. As a result, this technique is less laborious and gives the result in a much shorter period of time. Moreover, this method is quite sensitive enough to detect the presence of nondominant microbial communities in samples from previous years. However, there are certain limitations associated with the DGGE process. This includes lack of reproducibility as sometimes very high technical expertise is required, formation of primer dimers, and improper staining.

Terminal restriction fragment length polymorphism

Terminal restriction fragment length polymorphism (T-RFLP) involves the amplification of 16S rRNA gene, and further, these amplified products are subjected to restriction digestion by specific enzymes. These fluorescently labeled terminal restriction fragments (T-RFs) are then subjected to an automated sequencer. A fingerprint is generated on the basis of the variations in the T-RFs size, and the number of peaks generated is directly proportional to the number of T-RFs present. The relative abundance of different populations is calculated according to the height and peak of these fingerprints. For the analysis of obtained peaks, many

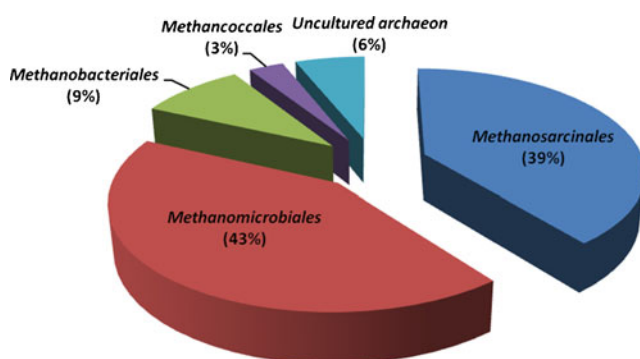


Fig. 1 Community structure composition of different orders of methanogens according to their dominance in sediments

software or online tools are available, such as TRF-CUT, TRiFle, and TAP-TRFLP. This method of T-RFLP has also been used by several groups to study the diversity of methanogens in sediments (Weber et al. 2001; Goffredi et al. 2008; Putkinen et al. 2009; Bhattarai et al. 2012). The advantage of using this method is its high sensitivity and reliability, and moreover, it is a high-throughput method for studying the dynamics, structure, and diversity of different microbial communities in a given sample (Nocker et al. 2007; Schutte et al. 2008). There are also certain disadvantages associated with the use of this way of deciphering diversity as this technique will only focus on the terminal fragments and, if the distantly related sequences have the same terminal restriction site, it will result in the same peak and thus become undistinguishable, and sometimes, the appearance of false or pseudo T-RFs is yet another drawback of T-RFLP.

Quantitative real-time PCR

Although the use of quantitative real-time PCR to study the relative counts of different microbes in sediment samples has been done for the past several years (Hristova et al. 2001; Miyashita et al. 2009; Savichtcheva et al. 2011; Kojima et al. 2012; Dang et al. 2013), very few reports have been published about the use of this method in the quantification and diversity analysis of methanogens in sediments particularly. Recently, Zeleke et al. (2013) have suggested the use of quantitative real-time PCR along with pyrosequencing to study the methanogen populations of an intertidal mudflat in the Yangtze River Estuary of China. This technique of quantitative real-time PCR involves real-time detection of a reporter molecule which will give an increased amount of fluorescence as there is an increase in the concentration of PCR product at the end of each cycle. This fluorescence value is measured as compared to the threshold value, and as a result, the relative abundance of different microbial communities in a particular sample is calculated. The standard curves for absolute quantification have been prepared from the known concentration of linearized plasmid samples or genomic DNA. Quantification can be performed by targeting taxon/order/species-specific genes. The only disadvantage of using this method is its high cost and inability to detect two or more samples within the same sample.

Temporal temperature gradient gel electrophoresis

Temperature gradient gel electrophoresis (TTGE) is similar to DGGE irrespective of the fact that, during this process, a temperature gradient was set up as compared to the denaturant gradient gel in the case of DGGE. The electrophoretic mobility of the DNA fragments decreases as denaturation

occurs. Till now, there is only one report known to us in which TTGE has been used to study the diversity of methanogens from sediment samples. Earl et al. (2005) have used TTGE for the analysis of 16S rDNA sequences to monitor the methanogen population present in the water and sediments of a small eutrophic lake in UK. The technique is found to be highly sensitive and detects even a single base pair change in the sequences and also has enormous potential for high-throughput screening.

Future techniques

Although a number of techniques have been used to decipher the diversity of methanogens from sediment samples, there also exist a huge space for some new experimental techniques to study it in a more detailed, specific, and less cumbersome way. Methanogen diversity has also been studied using some more specific techniques in some other systems, especially in ruminants. One of these techniques, which have not yet been used to study the methanogenic diversity in sediments, is amplified ribosomal DNA restriction analysis (ARDRA) technique. ARDRA is basically a fingerprinting technique which involves restriction digestion of the reamplified 16S rRNA sequences from the plasmid samples isolated from each clone. This procedure will help in the screening of a large number of samples for sequencing, and the unique pattern obtained by ARDRA helps in the identification or comparison of microbial communities (McSweeney et al. 2007). The first use of this technique has been done to study the spatial and temporal variations in bacterial marine plankton diversity (Acinas et al. 1997). Similarly, automated ribosomal intergenic spacer analysis, widely known as ARISA, was used. In this method, a fluorescence-labeled primer is used for the amplification of ribosomal intragenic region (Fisher and Triplett 1999). An automated laser detection system can detect the fluorescent DNA fragments during electrophoresis. Every band corresponds to at least one organism in the original community. So, ARISA might be further proven as a useful tool to study the diversity of methanogens in the sediments.

Suppressive subtractive hybridization (SSH) is another technique with huge potential to study methanogen diversity. SSH can rely on the identification of DNA fragments which is uniquely present in only one bacterial strain (Peng et al. 2011). The basis of this method is the comparison of the genomic content of bacterial species which are closely related to each other. In this technique, DNA fragments (strain-specific) is isolated and compared to the known gene or protein sequences, thus establishing a functional relationship between these DNA fragments and the compared genomes. SSH can be used as a novel approach for the comparative analyses of methanogenic communities present within the

sediments. Dot blot hybridization is yet another method which is quite easy and performed in a much shorter duration as compared to other lengthy techniques. This method is used to quantify the amount of specific 16S rRNA in a mixture relative to the total rRNA. This 16S rRNA is then hybridized with labeled oligonucleotide probes. McDonald et al. (2007) used dot blot hybridization analysis using class-level 16S rRNA gene probes as well as clone libraries to investigate the diversity of ultraphytoplankton during a specific period of time in the Gulf of Naples. Data obtained from the hybridization analysis showed the presence of at least three main classes: *Prymnesiophyceae*, *Cryptophyceae*, and *Chrysophyceae*. However, in addition, weaker signals from *Pelagophyceae* were also detected. All previously discussed techniques are still needed to study the diversity of methanogenic communities in sediments as there is much more need to understand the exact community structure of methanogens inhabiting inside these sediments. Only by understanding this scientists can make effective strategies to understand the methane emission process by these methanogens in the environment and can design further studies to deal with the ongoing problem of global warming.

Concluding remarks

Detailed understanding of the diversity of methanogens in sediments requires a thorough knowledge of the microbial community structure inside these sediments, such as how they are linked with methanogens, how the methanogen population depends on it, how they affect the substrate availability of the methanogens, and how they influence temperature by the production of different gases; these are all of the factors that we should understand deeply. Nucleic acid approaches allow the direct, reliable, and sensitive analysis of the microbial ecosystem inhabiting the sediments. If one has to study the diversity of any microbial community from these environments, then PCR cloning, sequencing, and DGGE will be the first choice. If somebody has any specific bacteria to search for or want to study the shift in the microbial communities, then probably he/she will go for probe-based methods or fingerprinting techniques like ARDRA, RFLP, etc. However, it has been found that not too many techniques have been used to study the methanogenic diversity from sediments and the diversity which is studied only up to the order level in most of the cases. So much more molecular approaches like ARDRA, ARISA, SSH, dot blot hybridization, etc. can be used to examine the diversity in these sediments which will help us in getting an exact picture of the types of methanogens found in different environmental conditions inside the sediments. These new and already used molecular approaches should be used in combination with traditional culture-based, microscopy, and microbiological techniques. These techniques, when used together, can give

us a much clearer view of methanogen diversity in the sediments and help scientists to make suitable efforts in the direction of methane mitigation from these sources to the environment and help in making a step forward to control the problem of global warming.

Acknowledgments The authors are thankful to the European Social Fund and state budget of the Czech Republic for providing the financial support during this study. This work is a part of the POSTUP II project CZ.1.07/2.3.00/30.0041, which is mutually financed by the previously stated funding agencies.

References

- Acinas SG, Rodríguez-Valera F, Pedros-Alio C (1997) Spatial and temporal variation in marine bacterioplankton diversity as shown by RFLP fingerprinting of PCR amplified 16S rDNA. *FEMS Microb Ecol* 24:27–40
- Antony CP, Murrell JC, Shouche YS (2012) Molecular diversity of methanogens and identification of *Methanolobus* sp. as active methylotrophic Archaea in Lonar crater lake sediments. *FEMS Microbiol Ecol* 81:43–51
- Bhattarai S, Ross KA, Schmid M, Anselmetti FS, Bürgmann H (2012) Local conditions structure unique archaeal communities in the anoxic sediments of meromictic Lake Kivu. *Microb Ecol* 64(2):291–310
- Bidre-Petit C, Jézéquel D, Dugat-Bony E, Lopes F, Kuever J, Borrel G, Viollier E, Fonty G, Peyret P (2011) Identification of microbial communities involved in the methane cycle of a freshwater meromictic lake. *FEMS Microbiol Ecol* 77:533–545
- Cahyani VR, Matsuya K, Asakawa S, Kimura M (2004) Succession and phylogenetic profile of methanogenic archaeal communities during the composting process of rice straw estimated by PCR-DGGE analysis. *Soil Sci and Plant Nut* 50:555–563
- Capone DG, Kiene RP (1988) Comparison of microbial dynamics in marine and freshwater sediments: contrasts in anaerobic carbon catabolism. *Limnol Oceanogr* 33:725–749
- Cetecioglu Z, Ince BK, Kolukirik M, Ince O (2009) Biogeographical distribution and diversity of bacterial and archaeal communities within highly polluted anoxic marine sediments from the Marmara Sea. *Mar Pollut Bull* 58:384–395
- Chaudhary PP (2009) *Methanomicrobium* phylotype are the dominant methanogen phylotype in the Murrah buffaloes. *Lett Appl Microbiol* 48:386
- Chaudhary PP, Sirohi SK (2009) Dominance of *Methanomicrobium* phylotype in rumen (*Bubalus bubalis*) methanogens from India. *Lett Appl Microbiol* 49:274–277
- Chaudhary PP, Sirohi SK, Saxena J (2012) Diversity analysis of methanogens in rumen of *Bubalus bubalis* by 16S riboprinting and sequence analysis. *Gene* 493:13–17
- Chaudhary PP, Sirohi SK, Singh D, Saxena J (2011) Methyl coenzyme M reductase (*mcrA*) gene based phylogenetic analysis of methanogens population in Murrah buffaloes (*Bubalus bubalis*). *J Microbiol* 49:558–561
- Cicerone RJ, Oremland RS (1988) Biogeochemical aspects of atmospheric methane. *Global Biogeochem Cy* 1:61–86
- Conrad R (2009) The global methane cycle: recent advances in understanding the microbial processes involved. *Environ Microbiol Rep* 1:285–292
- Dang H, Zhou H, Yang J, Ge H, Jiao N, Luan X, Zhang C, Klotz MG (2013) Thaumarchaeotal signature gene distribution in sediments

- of the northern South China Sea: an indicator of the metabolic intersection of the marine carbon, nitrogen, and phosphorus cycles? *Appl Environ Microbiol* 79:2137–2147
- DeLong EF (1992) Archaea in coastal marine environments. *Proc Natl Acad Sci USA* 89:5685–5689
- Dhillon A, Lever M, Lloyd KG, Albert DB, Sogin ML, Teske A (2005) Methanogen diversity evidenced by molecular characterization of methyl coenzyme M reductase A (*mcrA*) genes in hydrothermal sediments of the Guaymas Basin. *Appl Environ Microbiol* 71:4592–4601
- Earl J, Pickup RW, Ritchie DA, Edwards C (2005) Development of temporal temperature gradient electrophoresis for characterising methanogen diversity. *Microb Ecol* 50:327–336
- Ehhalt D, Prather M (2001) Atmospheric chemistry and greenhouse gases. In: Dentener F, Derwent R, Dlugokencky E, Holland E, Isaksen I, Katima J, Kirchhoff V, Matson P, Midgley P, Wang M (eds) *Climate change: the scientific basis*. Cambridge University Press, Cambridge, pp 239–287
- Fisher MM, Triplett EW (1999) Automated approach for ribosomal intergenic spacer analysis of microbial diversity and its application to freshwater bacterial communities. *Appl Environ Microbiol* 65:4630–4636
- Forster P, Ramaswamy V, Artaxo P, Bernsten T, Betts R, Fahey DW, Haywood J, Lean J, Lowe DC, Myhre G, Nganga J, Prinn R, Raga G, Schulz M, Dorland RV (2007) Changes in atmospheric constituents and in radiative forcing. In: Solomon S, Qin D, Manning M, Chen Z, Marquis M, Averyt KB, Tignor M, Miller HL (eds) *Climate change 2007: the physical science basis*. Contribution of Working Group I to the Fourth Assessment Report of the Intergovernmental Panel on Climate Change. Cambridge University Press, Cambridge
- Ganzert L, Jurgens G, Münster U, Wagner D (2007) Methanogenic communities in permafrost-affected soils of the Laptev Sea coast, Siberian Arctic, characterized by 16S rRNA gene fingerprints. *FEMS Microbiol Ecol* 59:476–488
- Garcia JL (1990) Taxonomy and ecology of methanogens. *FEMS Microbiol Rev* 87:297–308
- Giovannoni SJ, Britschgi TB, Moyer CL, Field KG (1990) Genetic diversity in Sargasso Sea bacterioplankton. *Nature* 345:60–63
- (2007) Global warming potentials. Contribution of Working Group I to the Fourth Assessment Report of the Intergovernmental Panel on Climate Change
- Goffredi SK, Wilpiseski R, Lee R, Orphan VJ (2008) Temporal evolution of methane cycling and phylogenetic diversity of archaea in sediments from a deep-sea whale-fall in Monterey Canyon, California. *ISME J* 2:204–220
- Grosskopf R, Stubner S, Liesack W (1998) Novel euryarchaeotal lineages detected on rice roots and in the anoxic bulk soil of flooded rice microcosms. *Appl Environ Microb* 64:4983–4989
- Hristova KR, Luttenegger CM, Scow KM (2001) Detection and quantification of methyl tert-butyl ether-degrading strain PM1 by real-time TaqMan PCR. *Appl Environ Microbiol* 67:5154–5160
- Ikenaga M, Asakawa S, Muraoka Y, Kimura M (2004) Methanogenic archaeal communities in rice roots grown in flooded soil pots: estimation by PCR-DGGE and sequence analyses. *Soil Sci and Plant Nut* 50:701–711
- Jiang L, Zheng Y, Chen J, Xiao X, Wang F (2011) Stratification of Archaeal communities in shallow sediments of the Pearl River Estuary, Southern China. *Anton Van Leeuwenhoek* 99:739–751
- Kanokratana P, Chanapan S, Pootanakit K, Eurwilaichitr L (2004) Diversity and abundance of Bacteria and Archaea in the Bor Klueng Hot Spring in Thailand. *J Basic Microbiol* 44:430–444
- Karr EA, Ng JM, Belchik SM, Sattley WM, Madigan MT, Achenbach LA (2006) Biodiversity of methanogenic and other archaea in the permanently frozen Lake Fryxell, Antarctica. *Appl Environ Microbiol* 72:1663–1666
- Keyser M, Witthuhn RC, Lamprecht C, Coetzee MP, Britz TJ (2006) PCR based DGGE fingerprinting and identification of methanogens detected in three different types of UASB granules. *Syst Appl Microbiol* 29:77–84
- Kojima H, Tsutsumi M, Ishikawa K, Iwata T, Mußmann M, Fukui M (2012) Distribution of putative denitrifying methane oxidizing bacteria in sediment of a freshwater lake, Lake Biwa. *Syst Appl Microbiol* 35:233–238
- Kumar S, Dagar SS, Mohanty AK, Sirohi SK, Puniya M, Kuhad RC, Sangu KP, Griffith GW, Puniya AK (2011) Enumeration of methanogens with a focus on fluorescence in situ hybridization. *Naturwissenschaften* 98:457–472
- Lazar CS, Dinasquet J, L'Haridon S, Pignet P, Toffin L (2011) Distribution of anaerobic methane-oxidizing and sulfate-reducing communities in the G11 Nyegga pockmark, Norwegian Sea. *Anton Van Leeuwenhoek* 100:639–653
- Lazar CS, John Parkes R, Cragg BA, L'Haridon S, Toffin L (2012) Methanogenic activity and diversity in the centre of the Amsterdam Mud Volcano, Eastern Mediterranean Sea. *FEMS Microbiol Ecol* 81:243–254
- McDonald SM, Sarno D, Scanlan DJ, Zingone A (2007) Genetic diversity of eukaryotic ultraphytoplankton in the Gulf of Naples during an annual cycle. *Aquat Microb Ecol* 50:75–89
- McSweeney CS, Denman SE, Wright ADG, Yu Z (2007) Application of recent DNA/RNA-based techniques in rumen ecology. *Asian-Aust J Anim Sci* 20:283–294
- Miyashita A, Mochimaru H, Kazama H, Ohashi A, Yamaguchi T, Nunoura T, Horikoshi K, Takai K, Imachi H (2009) Development of 16S rRNA gene-targeted primers for detection of archaeal anaerobic methanotrophs (ANMEs). *FEMS Microbiol Lett* 297:31–37
- Munson MA, Nedwell DB, Embley TM (1997) Phylogenetic diversity of Archaea in sediment samples from a coastal salt marsh. *Appl Environ Microbiol* 63:4729–4733
- Muyzer G, de Waal EC, Uitterlinden AG (1993) Profiling of complex microbial populations by denaturing gradient gel electrophoresis analysis of polymerase chain reaction amplified genes coding for 16S rRNA. *Appl Environ Microbiol* 59:695–700
- Muyzer G, Smalla K (1998) Application of denaturing gradient gel electrophoresis (DGGE) and temperature gradient gel electrophoresis (TGGE) in microbial ecology. *Anton van Leeuwenhoek* 73:127–141
- Nedwell DB (1984) The input and mineralization of organic carbon in anaerobic aquatic sediments. *Adv Microb Ecol* 7:93–130
- Nocker A, Burr M, Camper AK (2007) Genotypic microbial community profiling: a critical technical review. *Microb Ecol* 54:276–289
- Nolla-Ardévol V, Strous M, Sorokin DY, Merkel AY, Tegetmeyer HE (2012) Activity and diversity of haloalkaliphilic methanogens in Central Asian soda lakes. *J Biotechnol* 161:167–173
- Orphan VJ, Jahnke LL, Embaye T, Turk KA, Pernthaler A, Summons RE, DES Marais DJ (2008) Characterization and spatial distribution of methanogens and methanogenic biosignatures in hypersaline microbial mats of Baja California. *Geobiology* 6:376–393
- Peng M, Smith AH, Rehberger TG (2011) Quantification of *Propionibacterium acidipropionici* P169 bacteria in environmental samples by use of strain-specific primers derived by suppressive subtractive hybridization. *Appl Environ Microbiol* 77:3898–3902
- Purdy KJ, Munson MA, Nedwell DB, Martin Embley T (2002) Comparison of the molecular diversity of the methanogenic community at the brackish and marine ends of a UK estuary. *FEMS Microb Ecol* 39:17–21
- Putkinen A, Juottonen H, Juutinen S, Tuittila ES, Fritze H, Yrjölä K (2009) Archaeal rRNA diversity and methane production in deep boreal peat. *FEMS Microbiol Ecol* 70:87–98
- Rastogi G, Barua S, Sani RK, Peyton BM (2011) Investigation of microbial populations in the extremely metal-contaminated Coeur d'Alene River sediments. *Microb Ecol* 62:1–13

- Rigby M, Prinn RG, Fraser PJ, Simmonds PG, Langenfelds RL, Huang J, Cunnold DM, Steele LP, Krummel PB, Weiss RF, Doherty SO, Salameh PK, Wang H J, Harth CM, Mühle J, Porter LW (2008) Renewed growth of atmospheric methane. *Geophys Res Lett* 35(22). doi:10.1029/2008GL036037
- Rulik M, Bednarik A, Mach V, Brablcova L, Buriankova I, Badurova P, Gratzova K (2013) Methanogenic system of a small lowland stream Sitka, Czech Republic. *Biomass Now-Cultivation Utilization Chapter 17*:395–426
- Savichtcheva O, Debroas D, Kurmayer R, Villar C, Jenny JP, Arnaud F, Perga ME, Domaizon I (2011) Quantitative PCR enumeration of total/toxic *Planktothrix rubescens* and total cyanobacteria in preserved DNA isolated from lake sediments. *Appl Environ Microbiol* 77:8744–8753
- Schutte UM, Abdo Z, Bent SJ, Shyu C, Williams CJ, Pierson JD, Forney LJ (2008) Advances in the use of terminal restriction fragment length polymorphism (T-RFLP) analysis of 16S rRNA genes to characterize microbial communities. *Appl Microbiol Biotechnol* 80:365–380
- Sharma A, Chaudhary PP, Sirohi SK, Saxena J (2011) Structure modeling and inhibitor prediction of NADP oxidoreductase enzyme from *Methanobrevibacter smithii*. *Bioinformation* 6:15–19
- Sirohi SK, Singh N, Dagar SS, Puniya AK (2012) Molecular tools for deciphering the microbial community structure and diversity in rumen ecosystem. *Appl Microbiol Biotechnol* 95:1135–1154
- Sirohi SK, Chaudhary PP, Singh N, Singh D, Puniya AK (2013) The 16S rRNA and *mcrA* gene based comparative diversity of methanogens in cattle fed on high fibre based diet. *Gene* 523:161–166
- Surakasi VP, Wani AA, Shouche YS, Ranade DR (2007) Phylogenetic analysis of methanogenic enrichment cultures obtained from Lonar Lake in India: isolation of *Methanocalculus* sp. and *Methanoculleus* sp. *Microb Ecol* 54:697–704
- Wani AA, Surakasi VP, Siddharth J, Raghavan RG, Patole MS, Ranade D, Shouche YS (2006) Molecular analyses of microbial diversity associated with the Lonar soda lake in India: an impact crater in a basalt area. *Res Microbiol* 157:928–937
- Watanabe T, Asakawa S, Nakamura A, Nagaoka K, Kimura M (2004) DGGE method for analyzing 16S rDNA of methanogenic archaeal community in paddy field soil. *FEMS Microbiol Lett* 232(2):153–163
- Weber S, Lueders T, Friedrich MW, Conrad R (2001) Methanogenic populations involved in the degradation of rice straw in anoxic paddy soil. *FEMS Microbiol Ecol* 38:11–20
- Woese CR, Kandler O, Wheelis ML (1990) Towards a natural system of organisms: proposal for the domains Archaea, Bacteria, and Eucarya. *Proc Natl Acad Sci USA* 87:4576–4579
- Yoshioka H, Maruyama A, Nakamura T, Higashi Y, Fuse H, Sakata S, Bartlett DH (2010) Activities and distribution of methanogenic and methane-oxidizing microbes in marine sediments from the Cascadia Margin. *Geobiology* 8:223–233
- Young A (1992) DoE Report CWM039A+B/92. Available at http://en.wikipedia.org/w/index.php?title=Landfill_gas_migration&oldid=525869338
- Zeikus JG, Winfrey MR (1976) Temperature limitation of methanogenesis in aquatic sediments. *Appl Environ Microbiol* 31(1):99–107
- Zelege J, Lu SL, Wang JG, Huang JX, Li B, Ogram AV, Quan ZX (2013) Methyl coenzyme M reductase (*mcrA*) gene-based investigation of methanogens in the mudflat sediments of Yangtze River Estuary, China. *Microb Ecol*. doi:10.1007/s00248-012-0155-2
- Zepp Falz K, Holliger C, Grosskopf R, Liesack W, Nozhevnikova AN, Müller B, Wehrli B, Hahn D (1999) Vertical distribution of methanogens in the anoxic sediment of Rotsee (Switzerland). *Appl Environ Microbiol* 65:2402–2408
- Zhou M, McAllister TA, Guan LL (2011) Molecular identification of rumen methanogens: technologies, advances and prospects. *Anim Feed Sci Technol* 166–167:76–86
- Zoetendal EG, Cheng B, Koike S, Mackie RI (2004) Molecular microbial ecology of the gastrointestinal tract: from phylogeny to function. *Curr Issues Intest Microbiol* 5:31–47